MIL-M-55167B(CR)
5 January 1988
SUPERSEDING
MIL-M-55167A(EL)
27 August 1971

MILITARY SPECIFICATION

MOUNTING MT-1029( )/VRC & MOUNTING MT-1898( )/VRC

This specification is approved for use within Communication-Electronics Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1.1 SCOPE

1.1 Scope. This specification covers two types of mountings used to shock mount tactical vehicular radio equipment. The mountings are designated as follows: (See 6.4.)

Mounting MT-1029( )/VRC
Mounting MT-1898( )/VRC

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. The following specifications and standards form a part of this specification to the extent specified herein. Unless otherwise specified, the issues of these documents shall be those listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, U.S. Army Communications-Electronics Command and Fort Monmouth, ATTN: AMSEL-ED-TO, Fort Monmouth, New Jersey 07703-5000 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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SPECIFICATIONS

FEDERAL

QQ-S-571	Solder; Tin Alloy, Tin-Lead Alloy and Lead Alloy

MILITARY

MIL-P-116	Preservation; Methods of
MIL-V-173	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic, and Associated Electrical Equipment
MIL-T-713	Twine and Tape, Lacing and Tying (For use in Electrical and Electronic Equipment)
MIL-S-901	Shockproof Equipment, Class HI (High Impact), Shipload
MIL-W-6858	Welding; Aluminum, Magnesium, Non-Hardening Steels or Alloys, and Titanium, Spot, Seam and Stitch
MIL-P-11268	Parts, Materials, and Processes used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Signal Equipment
MIL-I-17563	Impregnants for Aluminum, Copper, Iron, Magnesium and Zinc Alloy Castings
MIL-C-55181	Connectors, Plug and Receptacle Intermediate (Electrical Waterproof)
MIL-E-55585	Electronics Equipment and Parts, Packaging of

STANDARDS

FEFERAL

FED-STD-151	Metals; Test Methods

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-276	Impregnation of Porous Nonferrous Metal Castings
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-810	Environmental Test Methods
MIL-STD-2073-2	Packaging Requirement Codes

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2.1.2 Government drawings. The following Government drawings form a part of this specification to the extend specified herein. Unless otherwise specified, the issues shall be those in effect on the date of the solicitation.

DRAWINGS

ELECTRONICS COMMAND

SC-A-46439	List of Accessories for Package Tester
SM-D-415990	Mounting MT-1029( )/VRC Assembly
SC-B-415045	Mounting Hardware Kit for MT-1029( )/VRC
SM-D-415051	Mounting MT-1898( )/VRC Assembly
SC-B-415087	Mounting Hardware Kit for MT-1898( )/VRC

(Copies of specifications, standards and drawings required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein (except for associated detail specification, specification sheets or MS standards), the test of this specification shall take precedence. Nothing in this specification, however, shall supersede applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Description.

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3.1.1 Mounting MT-1029( )/VRC. This mounting provides for vehicular shock mounting of Receiver-Transmitters RT-246( )/VRC and RT-524( )/VRC and Amplifier Power-Supply Group OA-3633( )/PRC-25. It contains a gasket sealed junction box which provides connections for distributing power, control, and signal voltages between the receiver-transmitter and other equipment in the vehicular radio set system. Two clamps which lock and secure the mounted equipment to the top tray are provided. A bag of mounting hardware is furnished with the mounting.

3.1.1.2 Mounting MT-1898( )/VRC. This mounting provides for vehicular shock mounting of Receiver, Radio R-442( )/VRC. It contains a gasket sealed junction box which provides connections for distributing power, control and signal voltages between the receiver and other equipment in the vehicular radio system. A clamp which locks and secures the mounted receiver is provided. A bag of mounting hardware is furnished with the mounting.

3.2 Construction. The mountings shall be constructed in accordance with the applicable drawing indicated below. (See 4.14.)

Mounting	Drawing
MT-1029( )/VRC	SM-D-415990
MT-1898( )/VRC	SM-D-415051

3.3 First article samples. The contractor shall furnish three first article samples of the mountings for approval, provided the invitation for bids require samples.

3.4 Marking. Marking shall conform to MIL-M-13231. External connector identification symbols markings shall be Group I as described in that specification. (See 4.4.)

3.4.1 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.4.2 Serial numbers. The mountings shall have serial numbers.

3.5 Finish. The mountings shall be finished in accordance with MIL-F-14072 and the equipment drawings. (See 4.4.)

3.6 Cleaning.

3.6.1 Parts. After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted. (See 3.19.)

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3.6.2 Units. After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.7 Riveting. Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. After riveting, the joints shall be tight, and joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces. (See 3.19.)

3.8 Securing of parts. (See 3.19.)

3.8.1 General. Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the equipment is subjected to specified bounce and shock tests. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.8.2 Securing of parts by threaded fasteners. Nylok or other screws with plastic devices, and Loctite or similar sealants shall not be subjected to temperatures in excess of 250°F (as during baking of paint). Loctite or similar sealants shall be applied in accordance with the manufacturer's instructions.

3.9 Soldering. (See 3.19.)

3.9.1 Solder. Solder shall conform to QQ-S-571. For soldering electrical connections, composition SN60, type WRMAP2 shall be used.

3.9.2 Acid or acid salts. No acid or acid salts shall be used in preparation for or during soldering, however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.9.3 Process. There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated, or underheated. The minimum necessary amount of flux and solder

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shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.10 Welding. Spot welds shall conform to MIL-W-6858. When spot or other intermittent welds are used to hold a part, the numbers of welds shall be at least two. (See 3.19.)

3.10.1 Cleaning prior to welding. Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material.

3.10.2 Process. Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slag inclusions, and other harmful defects.

3.10.3 Cleaning after welding. Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.11 Wiring and cabling. Wiring and cabling shall be neat and sturdy. (See 3.19.)

3.11.1 Cabling. Insulated wires shall be formed into cables as specified in the unit drawings. The cabling of wires shall be affected by tape or twine conforming to MIL-T-713. Individual conductors thus secured shall lie essentially parallel; however, this does not prohibit the use of twisted pairs.

3.11.2 Slack. Wires and cables shall be as short as practical except that sufficient slack shall be provided for the following purposes:

(a) To prevent undue stress on cable forms, wires, and connections, including connections to resiliently supported parts.

(b) To enable parts to be removed and replaced during servicing without disconnecting other parts.

(c) To provide for at least two replacements of the part to which the wire or cable is connected.

(d) To facilitate field repair at broken or cut wires.

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3.11.3 Protection. Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulation material.

3.11.4 Support. Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminal and undue change in position of the wire or cable (1) during and after subjection of the equipment to specified service conditions or (2) after service or repair of the equipment in a normal manner. Twine or tape shall not be used for securing wire and cable.

3.11.5 Splicing. Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.11.6 Connections. Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and non-metallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.12 Service conditions. The equipment shall meet the following service conditions where a test is referenced, meeting the test shall be considered as compliance with the requirements:

3.12.1 Temperature. (See 4.6.1.)

(a) Operating - Ambient temperature in the range of 145°F to -40°F (the 145°F temperature includes effect of sun-load). Exposure at the low temperature extreme not to exceed 72 hours, at any one time.

(b) Non-operating - Exposure in the range of 160°F to -80°F; exposure at the low temperature extreme not to exceed 24 hours, at any one time.

3.12.2 Humidity (moisture resistance). During testing as specified in 4.6.2, the equipment shall meet the following requirements:

(a) Paragraph 4.7 Continuity and short tests, at all operational steps in Method 507, Procedure I of MIL-STD-810.

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(b) The mounting shall show no evidence of deterioration, corrosion, or change in tolerance limits when inspected at the appropriate steps in Method 507, Procedure I of MIL-STD-810.

3.12.3 Immersion. The equipment shall be capable of meeting the conditions and test outlined in paragraph 4.6.3.

3.12.4 Salt spray. After testing as specified in paragraph 4.12, the mounting finish when examined with the unaided eye shall not exhibit white corrosion products visible through the chromate film or on accidental scratches through the chromate film. White corrosion products at sharp edges shall not constitute failure.

3.12.5 Vibration, bounce, and shock. As described in 3.13.

3.13 Vibration, bounce, and shock.

3.13.1 Vibration, equipment (see 4.6.4). The natural frequency of the equipment mounted on its shock mounts, and the transmissibility capabilities shall be as specified below. There shall be no secondary resonant conditions outside the natural frequency range.

(a) Before bounce and ballistic shock tests.

Mounting	Plane	Natural Frequency (Cps)	Transmissibility (not greater than)
MT-1029( )/VRC	Vertical	25-35	4:1
	Horizontal (front to rear)	14-35	5:1
	Horizontal (side to side)	18-35	3:1
MT-1898( )/VRC	Vertical, front	25-37	4:1
	Vertical, rear	21-35	4:1
	Horizontal (front to rear)	12-35	4:1
	Horizontal (side to side)	12-35	6.7:1

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(b) After bounce and ballistic shock tests.

Mounting	Plane	Natural Frequency (Cps)	Transmissibility (not greater than)
MT-1029( )/VRC	Vertical	20-35	4:1
	Horizontal (front to rear)	12-35	5:1
	Horizontal (side to side)	14-35	3:1
MT-1898( )/VRC	Vertical, front	22-37	4:1
	Vertical, rear	21-35	4:1
	Horizontal (front to rear)	10-35	4:1
	Horizontal (side to side)	10-35	6.7:1

3.13.1.1 Vibration, individual isolators. The individual isolators shall meet the following requirements when tested in accordance with paragraph 4.6.4.1:

Mounting	Natural Frequency (cps)	Transmissibility Max at Natural Frequency
MT-1029( )/VRC	25-35	4.5:1
MT-1898( )/VRC	25-35	4:1

3.13.2 Bounce and shock. The equipment shall be capable of meeting the requirements of Table I.

Table I - Bounce and Shock

Paragraph Heading	Paragraph No.	Performance after test
Bounce	4.6.5	Specified performance (Note A) No physical damage
Shock; ballistic	4.6.6	Operable (Note A) Any physical damage shall be minor only.

Note A: The mounting shall meet specified performance for the Continuity and Short tests (4.7) and the Vibration test (3.13.1(b)) after being subjected to this test.

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3.14 Continuity and short tests (see 4.7). The junction box shall be wired in accordance with the applicable drawings and shall meet the tests specified in paragraph 4.7.

3.15 Mounting hardware. A bag of mounting hardware shall be furnished with each mounting in accordance with the applicable drawing listed below:

SC-B-415045 Mounting Hardware Kit for MT-1029( )/VRC
SC-B-415087 Mounting Hardware Kit for MT-1898( )/VRC

3.16 Interchangeability. Like units, assemblies, sub-assemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.8.) Individual items shall not be hand-picked for fit or performance. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.17 Technical literature. Technical literature shall be furnished if specified in the contract. (See 6.2.)

3.18 Air seal. When tested as specified in 4.10, the decrease in vacuum shall not exceed 0.01 pounds per square inch, per minute.

3.19 Workmanship. The equipment shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs herein: (See 4.9.)

3.2 Construction	3.9 Soldering
3.6 Cleaning	3.10 Welding
3.7 Reveting	3.11 Wiring and cabling
3.8 Securing of parts	

3.20 Impregnation of aluminum die casting. Aluminum die castings shall be impregnated in accordance with MIL-STD-276, using impregnating material type III per MIL-I-17563. Castings which have been impregnated shall be marked "IMP". The impregnation shall correct general seepage only and shall not be used to correct poor foundry techniques or significant porosity. (See 4.4.)

3.21 Connectors. Connectors, receptable (cable connectors) shall be in accordance with MIL-C-55181. (See 4.4.)

3.22 Tropicalization of material. Fungus-inert materials shall be used where practicable. Such materials are listed in requirement 4 of MIL-STD-454. Use of materials which are nutrients for fungi shall be subject to approval by the contracting officer; and when approved, the nutrient materials shall be cleaned in accordance with requirement 9 of MIL-STD-454 and treated as follows:

(a) These materials shall be treated in accordance with 3.11.8 of MIL-P-11268:

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Cotton thread and duck
Cork
Felt, hair or wool
Leather
Linen thread
Thermosetting plastic materials using cotton, linen, or wood-flour as a filler or base

(b) These materials shall be treated to be resistant to moisture and fungi, in accordance with 3.11.8 of MIL-P-11268 by a method which is approved by the contracting officer:

Cellulose, regenerated
Cotton, other than thread of duck
Jute
Linen, other than thread

(c) Nutrient materials other than those in (a) and (b) above shall be pretreated with a suitable fungcidal incredient, retaining the electrical and physical properties required and shall be such as to permit the equipment to pass the fungus test in 3.22.5 of this specification.

3.22.1 Treating materials. Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.22.2 Toxicity. Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenance of the equipment, or during use of the finished items when used for the purpose intended.

3.22.3 Flexibility. Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.22.4 Statement of treatment. When required to be submitted by the contract, the statement of treatment shall describe in detail the materials to be treated and the treating materials and processes that the contractor proposes to use. (See 6.2(g).)

3.22.5 Fungus. Equipment and assemblies shall show no evidence of viable fungus or corrosion* when subjected to 28 days of fungus test in accordance with paragraph 4.13.

*NOTE: Corrosion is defined as any visible degradation that can be attributed to flaky, pitted, blistered, or otherwise loosened finish or metal surface.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize either his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items must meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of assuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling in quality conformance does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to acceptance of defective material.

4.1.2 Contractor quality assurance. The contractor shall provide and maintain a means of determining product conformance in accordance with the requirements specified herein or elsewhere in the contract. The Government, at its option, may perform any evaluation deemed necessary to assure the adequacy of the means employed and the effectiveness of the contractor's quality assurance methods and systems.

4.1.3 Government verification (See 6.6). All quality assurance operations required of the contractor shall be subject to Government verification at scheduled or unscheduled intervals. Verification will consist of the following:

(a) Surveillance of the contractor's operations to determine that practices, methods, and procedures of the contractor's quality assurance system are being applied and are, in fact, accomplishing the objectives of total compliance of the product with the requirements of this specification and the contract.

(b) Government product inspections to determine compliance of the product with the specification requirements and the contract.

4.2 Classification of inspection. Inspection shall be classified as follows:

(a) First article inspection (does not include packaging). (See 4.3.)

(b) Inspection covered by subsidiary documents. (See 4.4.)

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(c) Quality conformance inspection of equipment before packaging (See 4.5).

(d) Quality conformance inspection of packaging. (See4.11.)

4.3 First article inspection. This inspection shall consist of the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B, and group C (except EVR). (See Tables IV, V, and VI, respectively). The first article inspection shall be performed in the order shown in Table II.

Table II - First article inspection

Inspection (For additional preproduction inspection see 4.3)	Rqr Para.	Insp. Para.
Sample #1		
Visual and mechanical	3.19	4.9
Interchangeability	3.16	4.8
Continuity and Short	3.14	4.7
Air seal	3.18	4.10
Immersion	3.12.3	4.6.3
Vibration (equipment)	3.13.1	4.6.4
Vibration (individual isolator)	3.13.1.1	4.6.4.1
Bounce	3.13.2	4.6.5
Shock, ballistic	3.13.2	4.6.6
Immersion (repeat)	3.12.3	4.6.3
Fungus	3.22.5	4.13
Sample #2		
Visual and mechanical	3.19	4.9
Interchangeability	3.16	4.8
Continuity and Short	3.14	4.7
Air seal	3.18	4.10
Temperature	3.12.1	4.6.1
Humidity	3.12.2	4.6.2
Sample #3		
Visual and mechanical	3.19	4.9
Interchangeability	3.16	4.8
Continuity and Short	3.14	4.7
Air seal	3.18	4.10
Salt spray (See Note 2)	3.12.4	4.12

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NOTE 1: Other first article inspection may precede, follow, or be interspersed between the non-damaging tests.

NOTE 2: The equipment shall be thoroughly washed, cleaned, dried and refurbished, if necessary, before preceding with subsequent tests.

4.4 Quality conformance inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the acceptance inspection before preparation for delivery:

Item	Where required
Marking	3.4
Finish	3.5
Impregnation of aluminum die castings	3.20
Connectors	3.21

4.5 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspection specified in 4.4. and 4.5.1 through 4.5.3.2. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records as an element of Government verification. In addition the Government - at its discretion - may perform all or part of the specified inspection, to verify the contractor's compliance with specified requirements. (See 6.5.) Test equipment for Government verification inspection shall be made available by the contractor.

4.5.1 Group A inspection. Equipment shall be assembled into discrete lots for quality conformance inspection, utilizing the criteria of MIL-STD-105 for determination of lot composition. Unless otherwise specified the lot size shall be determined from Table III as related to production rates necessary to meet the delivery schedules of the contract. The lot size shall be adjusted throughout the life of the contract as necessary to provide for changes in contract delivery schedule and production rates. Each unit of each lot of equipment shall be inspected for conformance to all the examinations and tests required in Table IV. The quality conformance of each lot shall then be subjected to an audit utilizing the procedures of MIL-STD-105 under the general inspection levels and the acceptable Quality Level (AQL) indicated in Table IV. Group A inspection shall be performed in any manner satisfactory to the Government except that the air seal (4.7) shall be last. (See 6.5.)

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Table III - Lot size

Production Rate	Size of Lot
500 or more per month	1 weeks production
51 to 499 per month	2 weeks production
0 to 50 per month	1 months production

Table IV - Group A inspection

Inspection	Req Para	Insp Para	Major	AQL	Minor
Visual and mechanical	3.19	4.9			
Holding clamps			1.0%		4.0%
Mounting hardware kit			1.0%		4.0%
Junction box assembly			1.0%		4.0%
Overall mounting assembly			1.0%		4.0%
Continuity and Short	3.14	4.7		1.0%	
Air seal	3.18	4.10		1.0%	

4.5.2 Group B inspection. This inspection including sampling shall conform to Table V and to procedures for special inspection levels of MIL-STD-105. Group B inspection shall be performed on production lots of product that have passed group A inspection. (See 4.5.1 for lot formation.)

4.5.2.1 Group B sampling plans. The Group B sampling plans, for the AQL listed in Table V, shall be as follows:

AQL	Inspection
1.0%	S-4

Table V - Group B inspection

Inspection	Rqr Para	Insp Para	AQL (See 4.5.2.1)
Interchangeability	3.16	4.8	1.0%

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4.5.3 Group C inspection. This inspection shall consist of the tests specified in Table VI and shall be performed on units of product that have been subjected to and passed group A and group B inspection requirements. The Vibration (individual isolator) inspection may be performed prior to Groups A and B. Sample units shall be selected in accordance with 4.5.3.1.

4.5.3.1 Sampling for group C inspection. Sample units shall be selected at random for each group C inspection listed in Table VI in accordance with the following:

4.5.3.1.1 Sub-group I. One unit from the first 25 units shall be tested. One unit from the next 100 units shall be tested. For the next 5,000 units, one unit out of every 200 units shall be tested. Thereafter one unit from each 500 units or fraction thereof shall be tested. When Vibration (individual isolator) inspection is performed prior to Groups A and B, sampling shall be as follows: For the first 20,000 (for MT-1898( )/VRC) or 25,000 (for MT-1029 ( )/VRC) production isolators, one isolator from each 200 isolators shall be tested. Thereafter one isolator from each 500 isolators or fraction thereof shall be tested.

4.5.3.1.2 Sub-group II. One unit from each two months production shall be tested. The first sample selected shall be at the start of the contract from first quality conformance inspection production lot.

4.5.3.1.3 Sub-group III. Four units from the first 25 production units shall be tested as follows:

(a) Bounce and shock tests are to be performed on one unit.

(b) Temperature and humidity tests are to be performed on a second unit.

(c) The fungus test is to be performed on a third unit.

(d) The EVR is to be performed on a fourth unit.

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Table VI - Group C inspection

Inspection	Rqr. Para.	Insp. Para.
Sub-Group I		
Vibration (equipment)	3.13.1	4.6.4
Vibration (individual isolator)	3.13.1.1	4.6.4.1
Immersion	3.12.3	4.6.3
Sub-Group II		
Salt Spray	3.12.4	4.12
Sub-Group III		
Bounce	3.13.2	4.6.5
Shock, ballistic	3.13.2	4.6.6
Temperature	3.12.1	4.6.1
Humidity	3.12.2	4.6.2
Fungus	3.22.5	4.13
Equipment Verification Review	3.2	4.14

4.5.3.2 Noncompliance. All quality conformance inspection shall be halted, including group A and group B inspections, upon the occurrence of any group C failure. The contractor shall immediately report in writing each group C failure occurrence, including details of the failure and characteristics affected. The contractor shall immediately investigate the cause of failure and further report the results of investigation and details of the proposed corrective action on the processes and materials, as applicable, and on all units of product which are manufactured under the same conditions and which the Government considers subject to the same failure. Reports shall be forwarded, with a copy of the Government procurement quality assurance representative to the responsible technical activity designated in the contract. After corrective action has been taken, additional sample units shall be subjected to group C inspection (all inspections, or the inspections which the sample failed, at the option of the technical activity) and group A and group B inspections may be reinstituted. However, final acceptance and shipment will be withheld until group C reinspection results have shown that the corrective action was effective and approved by the technical activity. (See "Delivery and Conditional Acceptance of Contract Items Subject to Group C Inspection" Provisions elsewhere in the contract.)

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4.5.4 Reinspection of conforming group B and group C sample units. Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A inspection after repair of all visible damage. (See 6.3.)

4.6 Service condition tests. The equipment vibration, bounce, and shock tests shall be conducted with the mountings loaded as follows:

(a) Mounting MT-1029( )/VRC shall be loaded with Receiver-Transmitter, Radio RT-246( )/VRC or Receiver-Transmitter, Radio RT-524( )/VRC.

(b) Mounting MT-1898( )/VRC shall be loaded with Receiver, Radio R-442( )/VRC.

4.6.1 Temperature Test. The equipment shall be subjected to the temperature tests of MIL-STD-810 as follows: (See 3.12.1)

(a) High temperature - Test Method 501.2, Procedure I, Hot, Induced and Procedure II, Basic hot, induced.

(b) Low Temperature - Test Method 502.2, Procedures I & II (Temperatures as given in 3.12.1.)

At all those steps in Test Methods 501 and 502 where operation of the equipment is designated the test for Continuity of paragraph 4.7 (as modified above) shall be performed and met. The mounting shall show no evidence of deterioration, corrosion, or change in tolerance limits when inspected at the appropriate steps of Methods 501 and 502.

4.6.2 Humidity (moisture resistance). The equipment shall be tested in accordance with Method 507, Procedure I of MIL-STD-810 Hot-humid (Cycle), non hazardous-normal. Prior to testing the equipment shall meet full specification performance and inspection requirements for the measurements and conditions specified in 3.12.2. Measurements shall be made during the periods shown on the figure for the cycle. Do not move equipment from the humidity chamber for measurements, and do not leave power on after measurements have been completed. No repair or replacement of parts shall be permitted. If the equipment fails to meet the specified performance during cycling, or fails subsequently, it does not pass the test. In addition, if the equipment fails to meet specified requirements after final conditioning and adjustment, it does not pass the test. (See 3.12.2.)

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4.6.3 Immersion test. The equipment, without connector covers, shall be immersed to a minimum depth of 3 feet of fresh water for 2 hours. Immediately prior to immersion, the temperature of the equipment shall be 40°F, or more, above the temperature of the water. The tank in which the equipment is immersed shall be of sufficient capacity to maintain the water within ±2°F of its initial temperature, or the temperature of the water shall be maintained within those limits by other means. After completion of the 2-hour period of immersion, the equipment shall be removed from the water and wiped dry on exterior surfaces. When the equipment is opened, there shall be no evidence of leakage. (See 3.12.3.)

4.6.4 Test for equipment vibration. The mounting, loaded as specified in 4.6 shall be tested as follows, to determine conformance to 3.13.1.

(a) Secure the equipment on its shock mounts to a vibration table that can be controlled within 10 percent of the specified amplitude. The equipment shall be mounted on its normal operating position with connections in place.

(b) Vibrate the equipment successively, in three mutually perpendicular directions, over a frequency range of 10 to 55 cps. The total excursion of the applied vibration shall be 0.030 inch.

(c) In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

(d) Measure amplitude of the applied vibration and the vibration of the equipment.

(e) Determine the natural frequency of the equipment; the frequency at which the ratio of the equipment vibration to the applied vibration is the greatest shall be considered the natural frequency. The transmissibility shall be defined at the ratio fo the amplitude of the equipment to the amplitude of the vibration table.

4.6.4.1 Test for individual isolator vibration. The isolators used in Mounting MT-1029( )/VRC shall be individually loaded with a 12 pound 8 ounce mass. The isolators used in Mounting MT-1898( )/VRC shall be individually loaded with a 6 pound mass. The center of gravity of each mass shall be directly on the center line of the top mounting hole of the isolators. The isolators, loaded as specified above shall be tested as follows, to determine conformance to 3.13.1.1.

(a) Secure the isolators (five for the MT-1029 and four for the MT-1898) to a vibration table. The isolators shall be mounted in their normal upright operating position.

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(b) The total excursion of the applied vibration (vibration table) shall be 0.030 ± .003 inches. The isolators shall be vibrated in the vertical direction only in a frequency range of 10 to 55 cps.

(c) Measure the amplitude of the applied vibration and the amplitude of the output vibration of the isolator.

(d) Determine the natural frequency of the isolators; the frequency at which the ratio of the isolator vibration to the applied vibration is the greatest shall be considered the natural frequency. The transmissibility shall be defined as the ratio of the output amplitude of the isolator to the applied amplitude of the vibration table.

4.6.5 Bounce test (3.13.2). The mounting, loaded as specified in paragraph 4.6 shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows.

(a) Secure the equipment to the 48- by 48-inch vehicular adapter plate, in the same manner as it would be mounted in a vehicle. The adapter place shall be free of cracks and broken welds, and all unused holes except the 2-inch breather holes shall be plugged. No part of the mounted surface shall be less than 7 inches from the edge of the adapter plate.

(b) Cover the tester bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

(c) Place the adapter plate on the bed of the package tester. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to safeguard personnel, provided that the fore-and-aft motion of the adapter plate against the backstop is not restrained.

(d) Operate the package tester, shafts in phase in the synchronous mode, for a total of 3 hours at 283 ± 1 rpm. Turn the adapter plate 90 degrees at the end of each 45 minutes, each time in the same direction.

(e) The mounting shall meet the requirements and tests of paragraph 3.13.2 after the test.

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4.6.6 Shock test; ballistic (See 3.13.2). The test shall be conducted on the "Shock Testing Machine for Light Weight Equipment" shown in MIL-S-901. The mounting, loaded as specified in paragraph 4.6 shall be secured in its normal operating position to the steel test plate by means of the same fasteners used for vehicular installation of the equipment. The test shall consist of a total of 9 blows: one each 1-foot blow, 3-foot blow, and 5-foot blow on the back, side, and top of the test plate. As an alternative to reorienting the test plate for the blows on the side of the plate, equivalent rotation of the equipment under test is permissible. The mounting shall meet the requirements and tests of paragraph 3.13.2 after the test.

4.7 Continuity and short tests. Electrical continuity of wiring shall be in accordance with 3.14 and verified as follows:

(a) Point to point (continuity).

(b) Ground connections (continuity)

(c) Short testing using 100V DC potential between all ungrounded points and between all ungrounded points and ground. (Diode circuit in MT-1029( )/VRC Box is not applicable.)

4.8 Inspection for interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirement of 3.16. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

(a) The location of the two guide pins and the 18 pin receptacle connector.

(b) The minimum inside width (dimension) of the equipment cavity.

(c) The height of the mounting clamp(s).

4.9 Visual and mechanical inspection. Equipment shall be examined for the defects listed in MIL-STD-252 and the requirements of 3.19.

4.10 Air seal test. For this test a dummy junction box cover containing provisions enabling performance of the test shall be used. The mounting as field transported shall be subjected to a vacuum of 1 pound per square inch (1 pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the junction box. The vacuum shall be valved-off and the interior pressure measured during the ensuing period of 1 minute. During this 1-minute period the decrease in vacuum shall not exceed .01 pound. The gage used for measurement of the vacuum shall be of such accuracy that a difference of the .01 pound can be determined readily. (See 3.18.)

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4.11 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

(a) First Article Inspection of Packaging.

(b) Quality Conformance Inspection of Packaging.

4.11.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.11.2 Quality Conformance Inspection of Packaging.

4.11.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.11.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.11.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VII, "PACKING INSPECTION PROVISIONS". Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality level of 4.0 percent defective.

Table VII - Packing inspection provisions.

No.	Characteristic	Method of Inspection
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual

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Table VI - Packing inspection provisions. (Cont'd)

No.	Characteristic	Method of Inspection
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

4.12 Salt spray test. The mounting(s) shall be subjected to the salt spray test in accordance with Federal Test Method Standard No. 151 for 96 hours. Chromate coatings shall be aged for a minimum of 24 hours prior to test. After the test the mounting(s) shall conform to 3.12.4.

4.13 Fungus. The equipment and assemblies shall be subjected to 28 days of the fungus test specified in Method 508 of MIL-STD-810. For gasket or other non-hermetically sealed equipments, the parts and materials of the chassis shall be sprayed. The equipment shall be subjected to the test in the open condition. (See 3.22.5.)

4.14 Equipment verification review. The contractor shall perform an Equipment Verification Review (EVR), consisting of a complete technical audit of the Equipment on order against the drawings cited for construction (para. 3.2). The EVR shall consist of the following:

(a) An audit to establish that "as built" equipment (unit, assembly, module, or part) is in accordance with the cited end product drawings.

(b) An audit to establish that "as built" configuration of unit, assembly, module or part meet the acceptance requirements (other than equipment specification performance) specified for each unit, assembly, module, or part. The audit shall record all acceptance test methods used together with resulting verification test data.

(c) The contractor shall compile an EVR report(s) containing the following information:

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(1) Identification of unit, assembly, module, or part and details proving compliance with 4.14(a) and 4.14(b) above.

(2) Discrepancies noted.

(3) Corrective action taken.

(d) The contractor shall make available to the Government copies of his EVR report(s) no later than fifteen (15) days prior to the submission of the first production unit or lot. The results of the EVR shall be subject to inspection by authorized Government personnel at the time the first unit or production lot is offered for acceptance. Government inspection will be to the depth and extent necessary to demonstrate that the "as built" hardware is in accordance with the cited drawings. Copies of the EVR reports shall be made available to the Government personnel during the inspection. The contractor shall provide the following:

(1) Segregation of units, assemblies, modules, and parts to permit reviewing personnel access for detailed inspection.

(2) As required, responsible personnel from each functional department available for discussions in their respective areas.

(3) Adequate administrative support for the EVR.

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-2073-2, Tables I through VII, coded as follows: 10-1-00-00-NS-X-ED.

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

6. NOTES

6.1 Intended use. The mountings are used as vehicular shock mounts for units of Radio Sets AN/VRC-12( ), AN/VRC-43( ) through 49. Mounting MT-1029( )/VRC is used for mounting of Receiver-Transmitters RT-246( )/VRC and RT-524( )/VRC, and Amplifier-Power Supply Group OA-3633( )/PRC-25. Mounting MT-1898( )/VRC is used for mounting of Receiver, Radio R-442( )/VRC.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

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(b) Type required.

(c) Level A or B preservation and packing (see section 5).

(d) When first article rough handling tests are not required (see section 4).

(e) Place of final inspection.

(f) Technical literature required. (See 3.17.)

(g) Submission of statement referenced in 3.22.4 as soon as possible after award of contract. This statement should be submitted to the contracting officer.

(h) When first article packaging inspection reports require Acquisition Activity approval prior to production unit packing.

6.3 Group C inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2.)

6.4 Nomenclature. The parenthesis in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: MT-1029/VRC. The contractor should apply for nomenclature in accordance with the applicable clause in the contract. (See 1.1.)

6.5 Verification inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.6 Environmental pollution preventive. Disposal methods of packaging materials, to prevent environmental pollution, are contained in MIL-HDBK-742, the material specification, or may be obtained from the preparing activity of the applicable material specification.

6.7 Subject term (key word) listing.

Amplifier Power Supply Group OA-3633/PRC-25
Mounting
Mounting MT-1029( )/VRC
Mounting MT-1898( )/VRC
Radio Set AN/VRC-12

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Receiver-Transmitter RT-524/VRC
Receiver-Transmitter RT-246/VRC
Receiver R-442/VRC

6.8 Changes from previous issue. Asterisks (or vertical lines) are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

Custodian:

ARMY-CR

Preparing Activity

ARMY-CR

Project No. 5820-A849

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